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Socioeconomic status and breast cancer incidence in California for different race/ethnic groups

Kathleen Yost[1,*], Carin Perkins[2], Richard Cohen[1], Cyllene Morris[1] & William Wright[2]
[1]*Public Health Institute, 1700 Tribute Road, Suite 100, Sacramento, CA 95815-4402, USA, E-mail: kyost@ccr.ca.gov;* [2]*California Department of Health Services, Cancer Surveillance Section, Sacramento, CA, USA (*Author for correspondence)*

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Abstract

Objective: The majority of research on breast cancer risk and socioeconomic status (SES) has been conducted for blacks and whites. This study evaluates the relationship between SES and breast cancer incidence in California for four race/ethnic groups.
Methods: Principal component analysis was used to create an SES index using 1990 Census data. Untracted cases were randomly allocated to census block groups within their county of residence. A total of 97,227 female breast cancer cases diagnosed in California between 1988 and 1992 were evaluated. Incidence rates and rate ratios (RRs) were estimated and a χ^2 test for trend across SES levels was performed.
Results: SES was positively related to breast cancer incidence, and this effect was stronger for Hispanics and Asian/others than for whites and blacks. Adjusting by SES did not eliminate the differences in breast cancer rates among race/ethnic groups. RR differences between the race/ethnic groups were greatest in the lowest SES category and attenuated with increasing SES. An increasing trend over SES was statistically significant for all race/ethnic groups. Including randomly allocated cases affected RR estimates for white women only.
Conclusions: Our results are consistent with similar findings for the Los Angeles area but differ from previous results for the San Francisco Bay area.

Introduction

Differences in breast cancer incidence rates have been observed in California between non-Hispanic white (white), non-Hispanic black (black), Hispanic, and Asian/other women [1]. The age-adjusted incidence rates for invasive breast cancer for white women in California from 1988 to 1993 were significantly higher ($p < 0.01$) than comparable rates for black, Asians/other, and Hispanic women [2]. Breast cancer incidence rates have generally been found to increase as socioeconomic status (SES) increases [3, 4] and differences in breast cancer incidence between white and black women are diminished after SES has been accounted for [4]. Little research has been published on the relationship between breast cancer incidence rates and SES for race/ethnic groups other than blacks and whites (*e.g.* Asian, Hispanic). Analyses of the effects of SES on breast cancer incidence that have considered multiple race/ethnic groups in California were for limited geographic regions [3, 5]. It is important to understand whether SES is responsible for some of the race/ethnic differences in breast cancer incidence in California.

SES has been associated with several health outcomes including cardiovascular disease, mortality from all causes, infant mortality, mental disorders, and some types of cancer [6, 7]. Furthermore, SES has been shown to be inversely related to breast cancer mortality; that is, as SES increases, mortality rates decrease and survival times increase [8–10]. A widely accepted definition of SES as a combination of occupation, education, and income is based on Max Weber's definition of social class as consisting of three domains: class, status, and power [7]. A variety of methods have been used to measure SES in epidemiologic research, both at the individual and aggregate levels. However, since cancer registries do not routinely collect individual-level SES data, aggregate-level census data must be used to approximate individual-level SES.

Median years of education [4, 11] or percentage of individuals over 25 years of age with a certain level of education (*e.g.* high school diploma, bachelor's degree) [10] have been used as aggregate-level measures of education. Median household income or percent below the poverty level are often used to represent income. However, some researchers [12] claim poverty level is a better measure than median household income, while others stress the importance of accounting for family size when quantifying income as an indicator of SES [7, 10, 13]. The proportion below the poverty level may be too restrictive due to the way poverty level is defined; thus, multiples of that variable (*e.g.* 125% of the poverty level) are often used [10]. Occupation is the least common indicator of SES in epidemiologic studies, possibly because it is more difficult to measure than other indicators such as education or income; approximately seven questions are needed to appropriately code occupation [7]. Occupation can be separated by job type, such as managerial/professional, homemaker, technical/sales/administration, skilled labor, and unskilled labor [13]. It is common to characterize occupational status more crudely as white- or blue-collar occupation or simply as being a member of the working class [12]. Aggregate-level indicators that measure occupation, education, and income are often used to represent SES, but indices that combine several of these indicators have also been used [3, 7, 9].

It has been suggested that, although overlaps among the three domains exist, education, income, and occupation may represent slightly different aspects of SES and at least two of these indicators should be used in epidemiologic studies involving SES [7]. Indicators of SES are highly correlated [9] and could result in multicollinearity in a multivariate analysis. Principal component analysis has been recommended [14] and used [9] to combine multiple indicators of SES into a single composite measure. This approach is ideal for SES, which is recognized as a composite variable rather than a latent variable. If multiple domains represent a composite variable, failing to include a domain can lead to a biased composite [14]; therefore, it is important to include measures of all three domains (education, income, and occupation) in a composite SES measure.

The objective of this study is to evaluate the relationship between SES and breast cancer incidence for race/ethnic groups in addition to blacks and whites. Data from the statewide, population-based California Cancer Registry and the 1990 US Census were used to examine this issue.

Materials and methods

Measure of SES

Information on SES for individual cancer patients is not currently reported to the California Cancer Registry; therefore, aggregate-level SES is used as a surrogate to approximate individual-level SES. SES information was aggregated at the census block group level. Census block groups represent smaller geographic areas than census tracts, and census block groups contain approximately one-fourth of the individuals in a census tract, or approximately 1000 individuals. SES measured at the census block group level is preferred over SES measured at the census tract level in epidemiologic studies because census block groups are more homogeneous with respect to SES parameters and they provide a closer approximation of SES characteristics measured at the level of the individual [10, 12, 15].

Data from the 1990 census were used to create a composite SES score for each census block group. Selection criteria for census block groups were total population greater than zero, a median income greater than zero, and population over 25 years of age greater than zero. The first criterion eliminated census block groups without any residents at the time of the 1990 census since these block groups would not contribute any information to the analysis. The second criterion was necessary to eliminate non-residential areas such as dormitories and prisons. The third criterion was necessary because the education indicator used in this analysis is defined for individuals over 25 years of age. There were 21,519 census block groups in California in the 1990 census. After applying the selection criteria, 20,919 (97.2%) census block groups were retained for analysis. Seven indicator variables were combined into a single SES index using principal component analysis (SAS PRINCOMP procedure) similar to that reported in Dayal *et al.* [9]. Dayal *et al.* used "median school years" and "percentage of high school graduates" as measures of education; "median income" and "percentage living below the poverty level" represented income; and "median rent" and "median house value" were used to adjust for cost-of-living differences. Our principal component analysis was slightly different in several respects. Primarily we felt that, by omitting the occupation indicator, a comprehensive measurement of SES would not be achieved. Bollen and Lennox [14] caution that excluding causal indicators of a composite variable could have serious repercussions; therefore, we included two measures of occupation: "proportion with a blue-collar job" and "proportion older than 16 in the workforce without a job". Dayal *et al.* used the "proportion below the poverty level'' in their analysis, but we felt that ''proportion below 200% of the poverty level'' was more discriminating and used that indicator in our analysis. The proportion with certain levels of education was measured on the 1990 census, but median school years was not; therefore, we used an education index developed by Liu *et al*. [3] that weights the proportion of people in a census block group with a given level of education by the number of years needed to attain that level of education.

Following the principal component analysis only one component was retained based on the Kaiser criterion (*i.e*. eigenvalue greater than 1.0) and a scree plot. This component accounted for 59.7% of the variance. This component (referred to herein as the SES index) is a composite variable that combines the following seven indicators (the correlations of each indicator with the SES index is in parentheses): education index (0.87), proportion with a blue-collar job (−0.70), proportion older than 16 in the workforce without a job (−0.68), median household income (0.85), proportion below 200% of the poverty level (−0.87), median rent (0.63), and median house value (0.78).

A principal component is a weighted linear combination of the original variables, and a component score is a specific value for a component calculated for a given sampling unit [16, 17], which in this case is the census block group. The indicator variables ''education index'', ''median income'', ''median house value'', and ''median rent'' are directly associated with high SES and had positive loadings on the SES component. The indicator variables ''proportion below 200% of the poverty level'', ''proportion blue-collar workers'', and ''proportion in workforce with no job'' are inversely related to SES and had negative loadings. Therefore, a larger component score in this analysis represents a higher SES level. Standardized component scores for the SES index for the 20,919 census block groups were sorted and categorized into quintiles.

Breast cancer cases

In-situ and invasive incident breast cancer cases were selected from the California Cancer Registry using the following criteria: female, diagnosed between 1988 and 1992, and at least 15 years of age. Cases were separated into four mutually exclusive race/ethnicity categories: non-Hispanic white (white), non-Hispanic black (black), Hispanic, and Asian/other. The Asian/other race category includes Asian, Pacific Islander, American Indian, and women of unspecified race/ethnicity. Of those in the Asian/other category, approximately 98% are either Asian or Pacific Islander and 2% are American Indian or of unspecified race/ethnicity. Cases with unknown race or age (n = 1081) were excluded. Hispanic ethnicity was based on information in the medical record and on surname. Women with race coded as white, black, or unknown with a last name (or maiden name, when present) on the 1980 US Census list of 12,497 Hispanic surnames were categorized as Hispanic. The use of surname was adopted to more accurately classify Hispanic ethnicity, which is usually underreported in medical records [18].

Cases were limited to diagnosis between 1988 and 1992 because the SES index and corresponding population estimates were identified at the census block group level using information from the 1990 census. It was considered reasonable to assume that the SES and population size of a given census block group would not change markedly 2 years before or 2 years after the census year. This is a reasonable assumption based on other research that showed no appreciable changes in aggregate SES variables measured at the census tract level over a 10-year period [19]. These criteria resulted in the selection of 97,265 cases. Population estimates by age, sex, and race/ethnicity for all census block groups in California were obtained by special request from the Bureau of the Census using mutually exclusive race/ethnic categories from the MARS (modified age, race, sex, and Spanish ethnicity) file.

Geocoding breast cancer cases to census block groups

Residential address at the time of diagnosis was geocoded to 1990 census block groups. We were unable to geocode 3425 (3.5%) of the 97,265 cases. The US census estimates percent urbanization for counties in California. We classified counties as urban and rural based on percent urbanization and found that untracted cases were more likely to reside in rural counties. Using the same definitions of urban and rural, we found that census block groups in rural counties have lower SES levels on average than census block groups in more urbanized counties (data not shown). Untracted cases are typically excluded from analyses involving aggregate SES data obtained from the census [5, 19–21]. However, we felt that excluding untracted cases could introduce selection bias. We resolved this by randomly allocating untracted cases to census block groups within their county of residence, which resulted in geocoding 100% of the cases. We were able to assign an SES level to 97,227 (99.96%) of the cases. The 38 cases that were not assigned an SES level resided in census block groups that were excluded due to a lack of necessary SES data. Therefore this analysis was based on 97,227 incident breast cancer cases residing in 20,919 census block groups. Of these cases, 80.2% were white, 5.5% were black, 9.3% were Hispanic, and 5.0% were Asian/other.

Statistical analysis

Age of cases and the population were categorized into three groups (15–49, 50–64 and 65 years or older). Cases and population estimates in a given census block group were assigned the SES level of that census block group. That is, each census block group has a specific SES level. A case was assigned the SES level of the census block where she resided at the time she was diagnosed with breast cancer. Population estimates for a specific census block group were also assigned the SES level of that block group. Population estimates were multiplied by 5 to account for the use of 5 years of incident breast cancer cases. Population estimates and breast cancer cases were stratified by age, race/ethnicity, and the SES index. Age-specific and age-adjusted incidence rates using 5-year age categories were calculated by directly standardizing to the 1970 US population. These rates will be referred to herein as *stratified rates*. Breast cancer incidence rates were also estimated using Poisson regression (SAS GENMOD procedure) adjusting for age, race/ethnicity, and SES. These rates will be referred to herein as *modeled rates*. Race/ethnicity and age were forced into the Poisson model and SES and two-way interaction terms were added in a stepwise manner to identify the best-fitting model. An ideal model was one with both parsimony and a non-significant deviance ($p > 0.05$) or a dispersion term (deviance/degrees of freedom) near 1.0 [22, 23].

Age- and race/ethnicity-specific rate ratios (RR) comparing the highest and lowest SES levels were calculated using both the stratified and modeling approaches. A χ^2 test for linear trend in the log incidence rates for breast cancer across levels of SES was performed for each race/ethnic group (SAS GENMOD procedure). The significance level for these trend tests was set at $\alpha = 0.05$ and no adjustments were made for multiple comparisons.

Results

Stratified rate ratios

The distribution of incident breast cancer cases among females 15 years or older and age-adjusted incidence rates by SES and race/ethnicity are presented in Table 1,

Table 1. Number of incident breast cancer cases[a], population estimates[b], and age-adjusted breast cancer incidence rates[c] (per 100,000) for women aged 15 or older by SES and race/ethnicity, California, 1988–1992

SES[d]	Incident cases and population estimates									
	White		Black		Hispanic		Asian/other		All races combined	
	Cases	Population	Cases	Population	Cases	Population	Cases	Population	Cases	Population
SES Hi	23,352	9,607,945	384	248,540	1092	894,210	1348	1,302,655	26,176	12,053,350
SES4	18,836	8,983,955	679	512,945	1483	1,493,590	1214	1,322,255	22,212	12,312,745
SES3	16,262	7,897,330	970	733,130	1791	2,136,120	969	1,229,960	19,992	11,996,540
SES2	13,237	6,294,775	1172	978,175	2110	3,111,955	748	1,014,815	17,267	11,399,720
SES Lo	6308	2,878,580	2123	1,513,770	2587	5,016,740	562	857,455	11,580	10,266,545
Total	77,995	35,662,585	5328	3,986,560	9063	12,652,615	4814	5,727,140	97,227	58,028,900

SES[d]	Age-adjusted incidence rates (per 100,00)			
	White	Black	Hispanic	Asian/other
SES Hi	218.3	177.8	153.9	117.2
SES4	191.7	167.7	135.8	105.8
SES3	182.5	171.6	122.5	92.2
SES2	175.6	146.4	104.9	86.4
SES Lo	171.3	145.7	84	70.9
Unadjusted by SES	192.7	155.4	108.5	96.4

[a] Data set includes 3425 untracted cases that were randomly allocated to census block groups in their counties of residence. Cases with unknown race/ethnicity or age were excluded.
[b] Population estimates from the 1990 census were multiplied by 5 to reflect 5 years of cancer incidence data.
[c] Age-adjusted to the 1970 US population.
[d] SES is measured as a composite index combining seven indicator variables.

which shows that failing to adjust by SES leads to an overestimation of incidence rates for women in the lower SES quintiles and an underestimation of incidence rates for women in the higher SES quintiles. Table 1 also shows that adjusting rates by SES does not eliminate the differences in rates between race/ethnic groups. The previously established relationship of increasing breast cancer incidence rates with increasing SES [3, 4] was evident in our results. Figure 1 shows race/ethnicity-specific RRs based on age-adjusted rates, where the lowest SES quintile is the reference group. The overall risk of developing breast cancer increases with SES in all four race/ethnic groups. The effect of increasing SES level on the RR is modest and similar for blacks and whites, but it is more marked for Asian/others and Hispanics. Figure 2 presents SES-specific RRs based on age-adjusted rates for each race/ethnic group, with Asian/other women serving as the reference category. At each level of SES, Asian/other women have the lowest rates, followed by Hispanic, black, and then white women. The disparity in RRs between race/ethnic groups is most pronounced in the lowest SES category. The RRs comparing Hispanic to Asian/other women increases slightly with increasing SES because breast cancer rates for Hispanics increase more sharply across levels of SES than rates for Asian/others. However, the RRs comparing white and black women to Asian/other women diminish with increasing SES because the rates for Asian/other women increase more dramatically with SES than the rates for whites and blacks. The data presented in Figures 1 and 2 suggest that SES modifies the relationship between race/ethnicity and breast cancer risk.

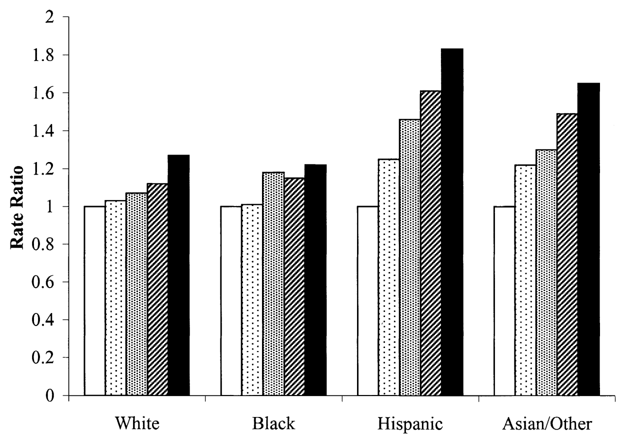

Fig. 1. Race-specific breast cancer rate ratios based on age-adjusted rates, women aged 15 or older, California, 1988–1992. Reference category is the lowest socioeconomic quintile (SES low). □, SES low; ⊡, SES2; ▦, SES3; ▨, SES4; ■, SES high.

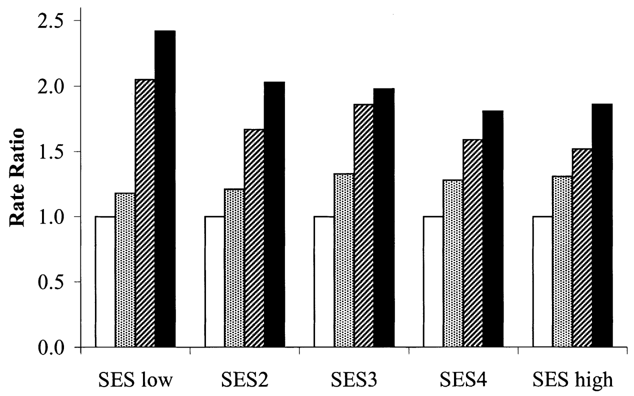

Fig. 2. SES-specific breast cancer rate ratios based on age-adjusted rates, women aged 15 or older, California, 1988–1992. Reference category is Asian/other. □, Asian/other; ▦, Hispanic; ▨, Black; ■, White.

Modeled rate ratios

The SES index created through principal component analysis improved the fit when entered into a Poisson regression model that contained only age and race/ethnicity. All two-way interactions between age, race/ethnicity, and SES were statistically significant and necessary for providing good fit to the data. This confirms that the effect modifications suggested from the evaluation of the *stratified* rates and RRs do exist and are statistically significant. The final Poisson regression model had a dispersion value of 1.166 and the p-value for the deviance was 0.26, both of which suggest the model fit the data well. *Modeled* age-specific RRs were calculated for each race/ethnic group and for each SES level with the lowest SES levels serving as the reference category (Table 2). *Modeled* RRs for Hispanic and Asian/other women are significantly greater than 1.0 ($p < 0.05$) for all of the SES quintiles. For whites and blacks the RRs for the second quintile (SES2) are not statistically significant, but all other RRs are. The effect of SES on breast cancer risk decreases with age for each SES level in Table 2. The decrease is most marked from the first to the second age group.

To evaluate the effect of including randomly allocated cases in our analysis we excluded these 3425 cases and recalculated the RRs. Excluding randomly allocated cases resulted in negligible changes in RRs for blacks, Asian/other, and Hispanics, but it markedly increased RRs for white women (data not shown). This is not surprising as most of the untracted cases were white (88.4%). Age-specific RRs comparing the highest and lowest SES quintiles were calculated with and without randomly allocated cases and were compared using the percent change in estimate [24]. The percent changes in

Table 2. Age, race/ethnicity, and SES specific breast cancer incidence rate ratios, estimated by Poisson regression[a], for women aged 15 years or older, California, 1988–1992

Race/ethnicity	Age group (years)	Rate ratios (95% confidence intervals) SES2	SES3	SES4	SES High	χ^2 test for trend[b]
Whites	15–49	1.04 (0.98–1.1)	1.14 (1.1–1.2)	1.31 (1.2–1.4)	1.78 (1.7–1.9)	<0.0001
	50–64	1.02 (0.97–1.1)	1.06 (1.0–1.1)	1.13 (1.1–1.2)	1.26 (1.2–1.3)	<0.0001
	65+	1.01 (0.97–1.1)	1.05 (1.0–1.1)	1.08 (1.0–1.1)	1.21 (1.2–1.3)	<0.0001
Blacks	15–49	1.01 (0.93–1.1)	1.25 (1.2–1.4)	1.32 (1.2–1.5)	1.70 (1.5–1.9)	0.026
	50–64	1.00 (0.92–1.1)	1.16 (1.1–1.3)	1.14 (1.0–1.3)	1.20 (1.1–1.4)	0.008
	65+	0.98 (0.91–1.1)	1.14 (1.1–1.2)	1.09 (1.0–1.2)	1.16 (1.0–1.3)	0.574
Hispanics	15–49	1.28 (1.2–1.4)	1.59 (1.5–1.7)	1.92 (1.8–2.1)	2.61 (2.4–2.8)	<0.0001
	50–64	1.26 (1.2–1.4)	1.48 (1.4–1.6)	1.67 (1.6–1.8)	1.85 (1.7–2.0)	<0.0001
	65+	1.24 (1.2–1.3)	1.46 (1.4–1.6)	1.59 (1.5–1.7)	1.78 (1.7–1.9)	<0.0001
Asian/others	15–49	1.26 (1.1–1.4)	1.41 (1.3–1.6)	1.74 (1.6–1.9)	2.26 (2.0–2.5)	0.0001
	50–64	1.24 (1.1–1.4)	1.31 (1.2–1.5)	1.51 (1.4–1.7)	1.61 (1.5–1.8)	0.0016
	65+	1.22 (1.1–1.4)	1.29 (1.2–1.4)	1.44 (1.3–1.6)	1.54 (1.4–1.7)	<0.0001

[a] Based on Poisson regression model adjusted for age, race/ethnicity, SES, and all two-way interactions. Reference category for RRs is the lowest quintile – SES low.
[b] *p*-Value for χ^2 test for trend in log rates across levels of SES.

estimates were less than 5% for all age-specific RRs for blacks, Hispanics, and Asian/others. The percent changes in estimates were 13.1%, 13.0%, and 8.9% for white women aged 15–49, 50–64, and 65+, respectively. A change in estimate of 10% or more is generally considered important in epidemiologic studies [24].

χ^2 *Test for trend*

A χ^2 test for linear trend in the log of the incidence rates for breast cancer across levels of SES was performed. SES was entered as a continuous (*i.e.* interval) variable for this analysis. The results of the trend test are presented in Table 2. Due to the statistically significant interactions between age and both race/ethnicity and SES, it was necessary to perform the trend test separately for each age group. As shown in Table 2, RRs increase over SES and the trends in the log incidence rates are statistically significant for all race/ethnic groups with the exception of blacks 65 years or older. The slopes of the log rates were greatest for Hispanics, followed by Asian/others, whites, and then blacks except for the 50–64 age group where the slope for blacks was greater than that for whites. The slopes of the log rates among the race/ethnic groups were not significantly different from one another in the 15–49 age group ($p = 0.195$), but they were significantly different in the 50–64 age group ($p = 0.017$) and the 65+ age group ($p < 0.0001$) indicating that differences by race/ethnicity remain following adjustment for SES. To ensure that including *in-situ* breast cancer cases was not influencing these results, we repeated the trend analysis without the 11,220 *in-situ* cases and were able to replicate our findings.

Discussion

Although previous research has evaluated the relationship between breast cancer and SES in different race/ethnic groups, the analyses were confined to limited geographic areas [3, 5] or to comparisons between blacks and whites [21]. This report presents the only analysis, to our knowledge, of the relationship between breast cancer incidence and SES across California in four race/ethnic groups using data from a population-based registry. We showed that, while SES is positively related to breast cancer incidence in all race/ethnic groups, the magnitude of the relationship is not the same for each group. When comparing breast cancer risk between groups of women, data should be adjusted for SES in addition to age and race/ethnicity, and interactions between these variables should be evaluated when assessing breast cancer risk via Poisson regression modeling.

Because of the interactions between the various components of the Poisson regression model, it was not possible to generate a summary RR for race/ethnic differences in breast cancer risk adjusted for SES. Nonetheless, SES- and race/ethnicity-specific age-adjusted rates indicate that SES explains only a portion of the race/ethnic differences in breast cancer risk. Based on age-adjusted rates in Table 1, breast cancer incidence unadjusted by SES was 78% higher among white women (192.7 per 100,000) than among Hispanic women (108.5 per 100,000). White women were at a higher risk than were Hispanic women in each SES category. Although the differences in rates generally decreased with increasing SES, rates for white women were still 41% higher than those for Hispanic women in the fourth SES category (191.7 and 135.8 per 100,000, respectively), where differences between whites and Hispanics were the smallest. The differences in rates between white and Asian/other women showed less change with SES, and SES had little influence on the difference in rates between white and black women.

We employed a random allocation method to assign untracted breast cancer cases to census block groups, enabling us to include those cases in our analysis. Most of the untracted cases (88.4%) were white. Untracted cases were more likely to reside in rural counties than tracted cases, and rural counties have lower SES on average than urban counties. Excluding untracted cases results in an overestimation of the effect of SES on breast cancer risk in white women, but results in negligible changes for other race/ethnic groups. The overestimation for whites was apparent when only 3.5% of breast cancer cases were not geocoded, which raises concerns about the common practice of excluding untracted cases from analyses. This is particularly troubling when one considers that the proportion of cases other researchers have excluded from analyses involving aggregate measures of SES are as high as 28% [19, 20]. Although we believe that including untracted breast cancer cases in our analysis yielded more accurate RRs, we must acknowledge the possibility that randomly allocating untracted cases to census block groups within their county of residence resulted in misclassification with respect to SES level, which in turn could have biased the RRs towards the null.

Despite this potential bias towards the null, we showed that breast cancer risk in California increases with SES in all race/ethnic groups, but that the magnitude of the effect is not the same for each race/ethnic group. In addition, race/ethnicity-specific incidence rates become somewhat more similar as SES increases, but controlling for SES does not eliminate race/ethnic differences in breast cancer risk. SES itself, whether measured at the individual or aggregate level, is not a causal factor for breast cancer. Rather, it may be a surrogate for individual-level breast cancer risk factors such as diet, lifestyle, age at first birth, parity, and duration of lactation [25, 26]. It may also be a surrogate for access to screening, which may account for some, but not all, of the SES effects observed in this study. Furthermore, the relationship between SES and risk factors or screening behavior may be different for each race/ethnic group. We conclude that the trend of increasing risk with SES is statistically significant for all race/ethnic groups except for blacks aged 65+. There are several possible explanations for this observation. As shown in Figure 1, an increase in RRs from the lower two SES levels to the three higher levels is apparent for black women, but the trend may not be linear in the log rates, particularly in the 65+ age group. The log rates for blacks do appear to increase slightly over levels of SES, but not as monotonically as they do for other race/ethnic groups (data not shown). The 65+ age group contains the largest proportion of blacks (35.1% of black cases); thus, the lack of significance in this age group is not likely to be due to insufficient power.

The results of our trend test and our results in general are consistent with an analysis done by Liu *et al.* in the Los Angeles area [3], but inconsistent with an analysis performed by Krieger *et al.* in the San Francisco Bay Area [5]. Both the Liu *et al.* and Krieger *et al.* analyses used population-based breast cancer data. Our method of measuring SES was more comparable to the method used by Liu *et al.* [3] than to that used by Krieger *et al.* [5]. Although Liu *et al.* did not perform race/ethnicity-specific trend tests on the log rates for breast cancer, their overall test for trend was highly significant ($p = 0.001$) and the relationships they observed between age-adjusted breast cancer incidence rates and SES levels (see their Table 3) are similar to what we observed with the following exceptions. The increase in breast cancer rates over SES levels for Hispanics reported by Liu *et al.* [3] appears to be slightly flatter than ours, and it is difficult to compare results for Asian/others because they evaluated them as separate race/ethnic groups. Krieger *et al.* [5] found a statistically significant increasing trend for Hispanics only, and while the trend for whites was not statistically significant, it was decreasing. That is breast cancer rates decreased as SES increased. There are several possible explanations for the discrepancy in our findings relative to Krieger *et al.* [5]. It is possible that the trends in the log rates for breast cancer across SES levels are apparent in some parts of the state (*e.g.* Los Angeles), but not others (*e.g.* San Francisco Bay Area). Other potential reasons include differences in the power of the trend test due to sample sizes and number of SES categories evaluated, differences in methods of measuring SES, differences in the treatment of untracted cases, inclusion of *in-situ* cases in our analysis, and differences in the extent of misclassification bias of race/ethnicity. Most of the misclassification of race/ethnicity in cancer registry data is related to undercounting of Hispanics. However, the magnitude of this misclassification is expected to be minimal due to methods employed by the registry to identify Hispanics based on surname. Stewart *et al.* [27] compared methods of classifying Hispanic ethnicity in a cancer registry in northern California. They calculated age-adjusted rates by ethnicity as reported to the registry and by ethnicity in which patients with surnames on the 1980 US census list of 12,497 Hispanic surnames were categorized as Hispanic. They found that correcting ethnicity based on surname resulted in a decrease in the age-adjusted breast cancer incidence rates from 153 to 150 per 100,000 for non-Hispanics and an increase from 76 to 93 per 100,000 for Hispanics. Both of the corrected rates were closer to estimates based on self-reported ethnicity of 151 and 91 per 100,000 for non-Hispanic and Hispanic breast cancer patients, respectively [27].

To examine the discrepancy between our results and those of Krieger *et al.*, we repeated our analysis but limited the breast cancer cases to those diagnosed in San Francisco Bay Area counties from 1988 to 1992. We excluded *in-situ* and randomly allocated cases to make our data set more similar to that used by Krieger *et al.* When the test for trend was performed on the resulting 17,275 cases with all age groups combined, it was statistically significant ($p < 0.001$) for each race/ethnic group except blacks ($p = 0.19$). The trend test was also repeated separately for each age group. A linear trend in the log rates was statistically significantly for all three age groups for Hispanics, two age groups for whites, two age groups for Asian/others, and no age groups in blacks. The slopes of the trends were positive in each race/ethnic group for all ages combined and for each age group when evaluated separately. Therefore, we believe that the disparate results between our statewide trend tests and the Krieger *et al.* [5] trend tests for the San Francisco Bay Area can be explained in large part by differences in statistical power. Furthermore, we believe the inverse relationship between SES and breast cancer risk observed by Krieger *et al.* in white women (*i.e.* negative slope) is due to their SES index, which may not yield a consistent ordinal measurement of SES.

Epidemiologic studies evaluating the relationship between cancer incidence and SES often focus on main effects, particularly age and race/ethnicity, without considering potential effect modification. Effect modification among several variables was observed in the *stratified* rates and RRs and confirmed as statistically significant interactions in a Poisson regression model. Race by age was the strongest interaction and had the greatest effect on the fit of the model and on the incidence rate and RR point estimates. This was followed by the race by SES interaction; the age by SES interaction was the weakest. The presence of these three interactions was also suggested in a previous analysis of breast cancer incidence rates for black and white women [21].

Our study has several limitations. The results presented here are specific to breast cancer incidence in California, and generalization to other cancer sites or geographic regions may not be warranted. SES as a quantifiable construct is still being defined and there are many ways to measure it [7]. In this study we chose to define SES as a composite variable that combines three generally accepted domains: education, income, and occupation. We also included measures of cost of living (*i.e.* median rent and median house value) to account for the diversity of living costs across California. It is possible that there are other components that should be included in a composite variable purporting to measure SES. For example, it has been previously reported that blacks pay higher taxes than whites on homes of equal value, and that whites tend to have greater economic returns for a given level of education compared to blacks, Hispanics, and American Indians [12]. This suggests that individuals in different race/ethnic groups with the same SES as measured by occupation, education, and income may not share the same level of power, prestige, and opportunity in their communities, and that including variables that measure these may improve the quantification of SES and could be an area for future research. A final limitation may be our use of aggregate-level SES to approximate individual-level SES. Aggregate and individual-level SES may have unique relationships to breast cancer incidence. Previous research has suggested that aggregate-level SES can either overestimate [11] or underestimate [19] individual-level SES effects. Aggregating SES over a small geographic area, such as the census block group, has been shown to approximate individual-level SES [15], but it may also have introduced area-level effects into the results for our study.

A standard approach for comparing breast cancer risk between groups of women includes adjusting for age and race/ethnicity to control for the potential effect modification and/or confounding that may be introduced by these variables. The results of our research show it is necessary also to adjust for SES when evaluating breast cancer risk. Because excluding even a small percentage of untracted cases can have an important impact on effect measures, cancer registries should increase their efforts to geocode cases to census block groups. Furthermore, different methods for allocating cases that cannot be geocoded to census block groups should be explored. Due to a lack of individual-level SES data in the cancer registries, aggregate-level SES is often used to approximate individual-level SES. Cancer registries should investigate the validity and reliability of SES variables currently available in medical records to determine whether this information could be used to measure SES at the individual level. Until individual-level data are available, however, efforts to improve aggregate-level data should continue, as should research on the validity of approximating individual-level SES with existing aggregate-level data.

Additional research is needed to understand what risk factors for breast cancer are associated with SES, and how exposure to those factors changes in different race/ethnic groups as SES changes. In particular, research is needed to understand the relationship between known breast cancer risk factors and SES for Hispanic and Asian/other women, since SES exhibits the strongest effect on breast cancer incidence rates in these groups.

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